Communication Protocol for the GL50EVO

Table of Contents

Revision	Date	Description
001	2015 / 03 / 27	New release
…	…	…
005	2016 / 02 / 04	major update, new structure
006	2016 / 02 / 08	Adding note: Boding code

1 Bluetooth

The GL50evo with Bluetooth Adapter is able to transmit the stored measurements via Bluetooth Low energy. The GL50evo always acts as GAP peripheral / GATT server.

Unless specified otherwise, all fields are in the order of LSO to MSO. Where LSO = Least Significant Octet and MSO = Most Significant Octet.

1.1 Services and Characteristics

1.1.1 Services

Specification Name	Assigned Number	UUID	Note
Generic Access	0x1800	00001800-0000-1000-8000-00805f9b34fb	
Generic Attribute	0x1801	00001801-0000-1000-8000-00805f9b34fb	
Glucose	0x1808	00001808-0000-1000-8000-00805f9b34fb	
Device Information	0x180A	0000180A-0000-1000-8000-00805f9b34fb	
Device Firmware Update	N/A	00001530-1212-efde-1523-785feabcd123	Only with new BLE Cap
Custom Service	N/A	0000fff0-0000-1000-8000-00805f9b34fb	Functions available since Meter FW v. x148 / x149

1.1.2 Service Characteristics

Glucose

Name	Assigned Number	Properties	Note
Glucose Measurement	0x2A18	Notify	
Glucose Measurement Context	0x2A34	Notify	
Glucose Feature	0x2A51	Read	
Record Access Control Point	0x2A52	Write, Indicate	

Device Information

Name	Assigned Number	Properties	Note
System ID	0x2A23	Read	Changed with new BLE Cap
Model Number String	0x2A24	Read	Removed with new BLE Cap
Serial Number String	0x2A25	Read	
Firmware Revision String	0x2A26	Read	Removed with new BLE Cap
Hardware Revision String	0x2A27	Read	Removed with new BLE Cap
Software Revision String	0x2A28	Read	
Manufacturer Name String	0x2A29	Read	Changed with new BLE Cap
PnP ID	0x2A50	Read	Changed with new BLE Cap

Device Firmware Update

Name	UUID	Properties	Note
DFU Packet	00001532-1212-efde-1523-785feabcd123	Write	
DFU Control Point	00001531-1212-efde-1523-785feabcd123	Write, Notify	
DFU Version	00001534-1212-efde-1523-785feabcd123	Read	

Custom Service

Name	**UUID**	**Properties**	**Note**
Custom Characteristic 1	0000fff6-0000-1000-8000-00805f9b34fb	Write	Functions available since Meter FW v. x148 / x149
Custom Characteristic 2	0000fff7-0000-1000-8000-00805f9b34fb	Read	Functions available since Meter FW v. x148 / x149
Custom Characteristic 3	0000fff8-0000-1000-8000-00805f9b34fb	Notify	Not in use
Custom Characteristic 4	0000fffc-0000-1000-8000-00805f9b34fb	Write	Not in use, Removed with new Cap

1.2 Glucose Service

To receive a measurement the Collector needs to perform a “Record Access Control Point” procedure.

Before performing any “Record Access Control Point” procedure, the Collector needs to subscribe the Client Characteristic Configuration for “Glucose Measurement”, “Glucose Measurement Context” and “Record Access Control Point”.

1.2.1 Glucose Measurement

A notification delivered through the “Glucose Measurement” characteristic contains the following information:

Name	**Format**	**Info**	**Note**
Flags	8bit	Define which data fields are present	
Sequence Number	uint16		
Base Time	7byte		
Time Offset	sint16	Always: (0x) 00 00	Removed with Meter FW v. x158 / x159
Glucose Concentration	SFLOAT	Unit: kg/l or mol/L	
Type	nibble	Always: (0x) 1	
Sample Location	nibble	Always: (0x) 1	
Sensor Status Annunciation	16bit	Always: (0x) 00 00	Removed with Meter FW v. x158 / x159

To get more detailed information about the meaning of the values and flags check the “Bluetooth Service Specification – Glucose Service”

1.2.2 Glucose Measurement Context

For GL50evo the “Glucose Measurement Context” is only be used to transmit the marker. If no marker is set for a measurement, no “Glucose Measurement Context” notification is delivered for this measurement.

A notification delivered through the “Glucose Measurement Context” characteristic contains the following information:

Name	**Format**	**Info**	**Note**
Flags	8bit	Always: (0x) 02	
Sequence Number	uint16		
Meal	uint8	The possible values are defined in the following table “Meal values”	

Meal values:

Value	**Marker**		**Bluetooth specification name**
(0x) 01		Before meal	Preprandial
(0x) 02		After meal	Postprandial

(0x) 04	✱	General label	Casual

1.2.3 Glucose Feature

A response of read the “Glucose Feature” characteristic contains the following information:

Name	Format	Info	Note
Flags	16bit	Always: (0x) ff 07	Changed with new BLE Cap, Now always: (0x) 41 01

To get more detailed information about the meaning of the flags check the “Bluetooth Service Specification – Glucose Service”

1.2.4 Record Access Control Point

The GL50evo starts to send measurements when the Collector writes a command with the desired action to the “Record Access Control Point” and all Client Characteristic Configuration are subscribed.

The format of “Record Access Control Point” command is as follows:

Name	Format	Info	Note
Op Code	uint8	The possible values are defined in the following table “Supported Op Code value”	
Operator	uint8	The possible values are defined in the following table “Supported Operators”	
Operand Filter Type	uint8	The possible values are defined in the following table “Supported Operand Filter Types”	
Operand Filter Parameters	variable	<minimum filter value><maximum filter value> Format if “Filter Type” = “User Facing Time”: “Base Time” + “Time Offset”	

Supported Op Codes:

Value	Type
(0x) 01	Report stored records
(0x) 04	Report number of stored records

Supported Operators:

Value	Type
(0x) 01	All records
(0x) 02	Less than or equal to
(0x) 03	Greater than or equal to
(0x) 04	Within range of (inclusive)
(0x) 05	First record(i.e. oldest record)
(0x) 06	Last record (i.e. most recent record)

Supported Operand Filter Types:

Value	Type
(0x) 01	Sequence Number
(0x) 02	User Facing Time

Notes:

- *Please find examples in section 1.4.1.*
- *The following combinations are **not** supported:*
    - *“Report stored records” --> “Less than or equal to” --> “User Facing time”*
    - *“Report number of stored records” --> “Within range of” --> “User Facing time”*

- *In case of*
  - *“Report stored records” --> “Within range of” --> “User Facing Time”*

  the command would be larger as the Maximum Transmission Unit (MTU). Therefore the <maximum filter value> must discard “Time Offset” (example 3 in section 1.4.1).

In case of “Report number of stored records” is send as Op Code and **no error** occurs, the “Record Access Control Point” delivers an indication which contains the following information:

Name	Format	Info	Note
Op Code	uint8	(0x) 05	
Operator	uint8	(0x) 00	
Operand	uint16	Containing number of records	

In case of “Report stored records” is send or an **error** occurs, the “Record Access Control Point” delivers an indication which contains the following information:

Name	Format	Info	Note
Op Code	uint8	(0x) 06	
Operator	uint8	(0x) 00	
Request Op Code	uint8	Contains the requested Op Code	
Response Code	uint8	The possible values are defined in the following table “Supported Response Codes”	

Supported Response Codes:

Value	Type
(0x) 01	Success
(0x) 02	Op Code not supported
(0x) 03	Invalid Operator
(0x) 04	Operator not supported
(0x) 05	Invalid Operand
(0x) 06	No records found
(0x) 07	Abort unsuccessful
(0x) 08	Procedure not completed
(0x) 09	Operand not supported

To get more detailed information about the “Record Access Control Point” characteristic check the “Bluetooth Service Specification – Glucose Service”

1.3 Custom Service

The “Custom Service” is used to
- Read the “Meter FW version”
- Read the “BLE Cap version”
- Read the “Meter HW version”
- Write the “BLE turned off” command
- Read the “Sending mode”

All “Custom Service” functions are available since “Meter FW version” x148 / x149.

To read one of these information you need to write the desired command to “Custom Characteristic 1”, wait for 1000ms and then read “Custom Characteristic 2”.

To write the “BLE turned off” command you need to write this command to “Custom Characteristic 1”.

Note: Immediately after the GL50evo receives this command it disconnects and stop sending advertising packages until the user trigger the next data transmission. To avoid automatic reconnects the Collector should use this command to disconnect.

The format of a “Custom Characteristic 1” command is as follows:

Name	Format	Value	Note
Meter FW version	uint8	(0x) 01 00 00 00 00 00 00 00	
BLE Cap version	uint8	(0x) 02 00 00 00 00 00 00 00	
Meter HW Version	uint8	(0x) 03 00 00 00 00 00 00 00	
BLE turned off	uint8	(0x) 04 00 00 00 00 00 00 00	
Sending mode	uint8	(0x) 05 00 00 00 00 00 00 00	

A response of read “Custom Characteristic 2” contains the following information:

For command	Format	Info	Note
Meter FW version	uint8	String value encoded with UTF-8	
BLE Cap version	uint8	String value encoded with UTF-8	
Meter HW Version	uint8	String value encoded with UTF-8	
Sending mode	uint8	The possible values are defined in the following table “Sending mode values”	

Sending mode values:

Value	Info
(0x) 00 00 00 00 00 00 00 00	User starts data transfer by entering the memory mode
(0x) 01 00 00 00 00 00 00 00	Data transfer starts after a measurement

Note: The “Sending mode” information could be used to get the last measurement in case the data transfer starts after a measurement was taken.

1.4 Miscellaneous

1.4.1 Record Access Control Point Samples

1. Report the last stored record

Byte Index	Values	Descriptions
#1	0x01	Report Stored Records
#2	0x06	Last Record

2. Report stored records within the range of serial number 4 to 11:

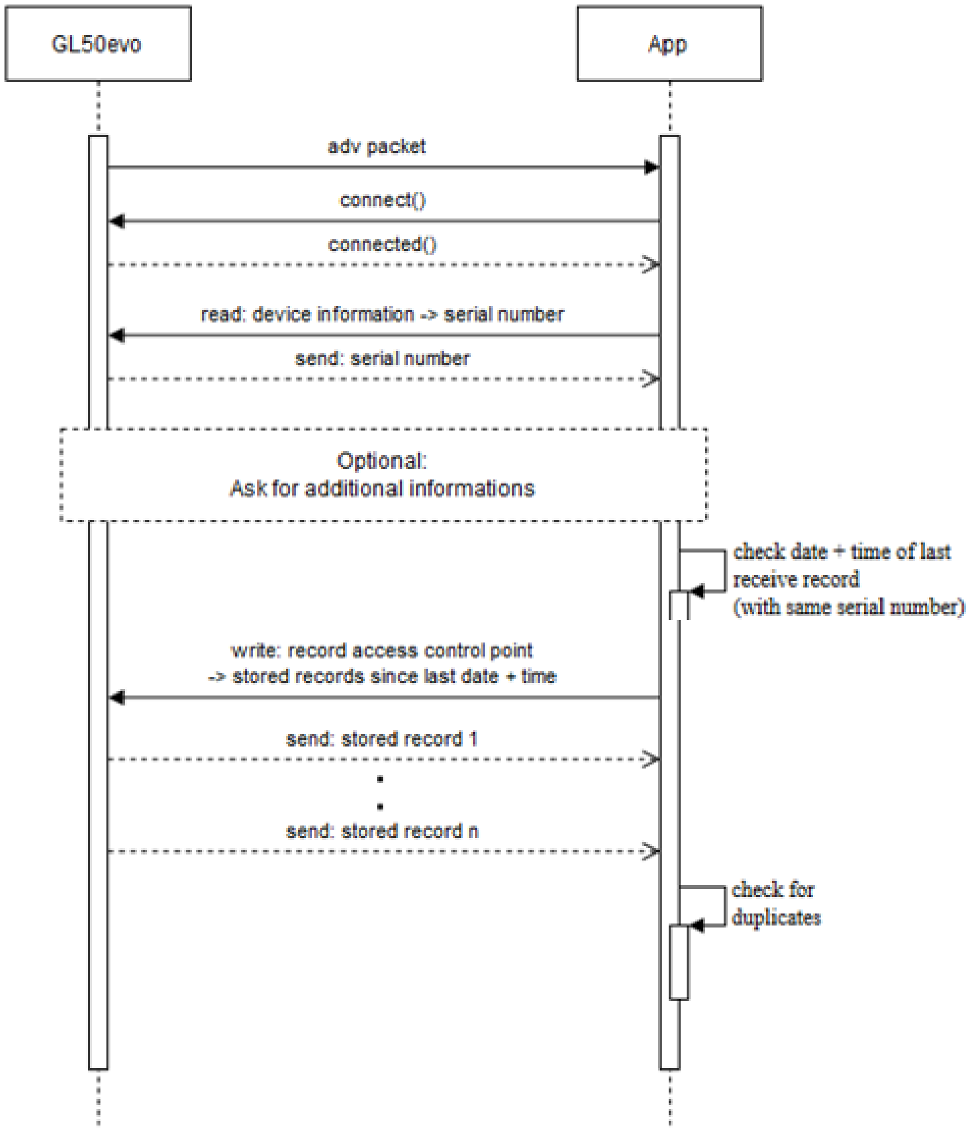
Byte Index	Values	Descriptions
#1	0x01	Report Stored Records
#2	0x04	Within range of (inclusive)
#3	0x01	Filter Type
#4	0x04	minimum filter value Low Byte
#5	0x00	minimum filter value High Byte
#6	0x0B	maximum filter value Low Byte
#7	0x00	maximum filter value High Byte

3. Report stored records within the range of 2015/02/01 12:59:00 to 2015/03/01 12:59:00

Byte Index	Values	Descriptions
#1	0x01	Report Stored Records
#2	0x04	Within range of
#3	0x02	Filter Type
#4	0xDF	Year Low Byte
#5	0x07	Year High Byte
#6	0x02	Month
#7	0x01	Day
#8	0x0C	Hour
#9	0x3B	Minute
#10	0x00	Second
#11	0x00	Time Offset Low Byte
#12	0x00	Time Offset High Byte
#13	0xDF	Year Low Byte
#14	0x07	Year High Byte
#15	0x03	Month
#16	0x01	Day
#17	0x0C	Hour
#18	0x3B	Minute
#19	0x00	Second

1.4.2 Proposal for data transmission

The app should not always ask for all 480 measurements because it will take a long time to transfer all 480 measurements. Beurer suggest that the app only asks for new measurements since the last data transfer:



Note: If the app supports multiple GL50evos, the app needs to save the serial number of the GL50evo for each measurement and ask for all measurements since the last measurement of the same GL50evo was transferred.
If this is ignored, it could happen that not all measurements are transferred to the app!

2 NFC

2.1 Records Format

Offset	***Meaning***	***Description***
0	(1st record) High byte of Glucose	BCD
1	(1st record) Low byte of Glucose	BCD
2	(1st record) Year	BCD
3	(1st record) Month	BCD
4	(1st record) Day	BCD
5	(1st record) Hour	BCD(24 hour format)
6	(1st record) Minute	BCD
7	(1st record) Type	Before lunch/After lunch/Marked
8	(2nd record) High byte of Glucose	BCD
9	(2nd record) Low byte of Glucose	BCD
10	(2nd record) Year	BCD
11	(2nd record) Month	BCD
12	(2nd record) Day	BCD
13	(2nd record) Hour	BCD(24 hour format)
14	(2nd record) Minute	BCD
15	(2nd record) Type	Before lunch/After lunch/Marked
…	…	…
8*479	(480th record) High byte of Glucose	BCD
8*479+1	(480th record) Low byte of Glucose	BCD
8*479+2	(480th record) Year	BCD
8*479+3	(480th record) Month	BCD
8*479+4	(480th record) Day	BCD
8*479+5	(480th record) Hour	BCD(24 hour format)
8*479+6	(480th record) Minute	BCD
8*479+7	(480th record) Type	Before lunch/After lunch/Marked
…	…	…
3874	Low byte of record numbers	binary
3875	High byte of record numbers	binary
3877	Low byte of record pointer	binary
3878	High byte of record pointer	binary
3960	For I2C Read/Write test	0xCA
4050	Model name	“GL50evo_NFC”
4076	F/W version	e.g.: ”0198”、”0199”
4084	Serial number	e.g.: “A13/001234”

Marker definition

Marker	***Code***
No marker	0x01
Before lunch	0x02
After lunch	0x03
Marked	0x04

Record pointer

8 bytes per record pointer

Device unit

Included in F/W version:

- XXX8 = mmol/L device
- XXX9 = mg/dL device

3 SCSI USB

3.1 Records Format

Offset	***Meaning***	***Description***
3874	Low byte of Record No(N)	
3875	High byte of Record No	
3877	Low byte of Base Address	
3878	High byte of Base Address	

Offset	***Meaning***	***Description***
0		
1		
2		
3	Reserved	
4	Reserved	
5	High byte of Glucose	BCD
6	Low byte of Glucose	BCD
7	Year	BCD
8	Month	BCD
9	Day	BCD
10	Hour	BCD (24 hour format)
11	Minute	BCD
12	Marker	01h: No marking, 02h: before meal 03h: after meal, 04h: general marking
…	…	…
N * 8-3	High byte of Glucose	BCD
N * 8-2	Low byte of Glucose	BCD
N * 8-1	Year	BCD
N * 8	Month	BCD
N * 8+1	Day	BCD
N * 8+2	Hour	BCD (24 hour format)
N * 8+3	Minute	BCD
N * 8+4	Marker	01h: No marking, 02h: before meal 03h: after meal, 04h: general marking
N * 8+5	Check Sum	0xff & Sum(byte0 + byte 1+…..+byte N*8+5)

3.2 Read Data Command

SCSI read (6) command

Byte	Name	Description
0	Operation code	08h
1	Reserved	
2	Logical Block Address (High Byte)	
3	Logical Block Address (Low Byte)	
4	Transfer Length	
5	Control	

Read Serial number

Logical Block Address (High Byte) should be set to 'S' and Logical Block Address (Low Byte) should be set to 'N'. Transfer length is 11 bytes, include 10 bytes serial number and 1 bytes check sum.

Read F/W version

Logical Block Address (High Byte) should be set to 'F' and Logical Block Address (Low Byte) should be set to 'V'. Transfer length is 3 bytes, including 2 bytes version number and 1 byte checksum.

Read H/W version

Logical Block Address (High Byte) should be set to 'H' and Logical Block Address (Low Byte) should be set to 'V'. Transfer length is 3 bytes, including 2 bytes version number and 1 byte checksum.

Read model name

Logical Block Address (High Byte) should be set to 'M' and Logical Block Address (Low Byte) should be set to 'N'. Transfer length is 8 bytes, including 7 bytes model name, "GL50evo" (unchangeable string), and 1 byte checksum.

4 Notes

a) All measurement values will be stored on device in mg/dL. NFC only transfers mg/dL values. Mmol/L value can be calculated by:

Value mg/dL
------------------ = Value mmol/L
18,0

Display on mmol/L device is with one decimal place, mathematical rounded.

b) All values <20mg/dL (<1.1mmol/L) will be displayed as “Lo” and should be displayed in software as well “Lo”

c) All values >630mg/dL (>35.0mmol/L) will be displayed as “Hi” and should be displayed in software as well “Hi”

d) Bluetooth on Android devices is sometimes on some devices not stable at initial connect
- Send connect command; if you detect a disconnect by your app, re-send connect command

e) For GL50evo under Meter FW Version x148 / x149 the app needs to ask 2 times for the first data packet. First command starts the bonding.

f) Since new BLE Cap user needs to enter a code for bonding the device with the smartphone.
Bonding Code: **982249** (The code is the same for all caps)

g) Depending on Bluetooth connection, the GL50evo may send single measurements twice:
Check received measurements for doublets (sequence number could be used for an easy check).

h) GL50evo will continue sending advertising packages as long as:
   a. Bluetooth cap is attached on GL50evo
   b. Bluetooth is activated in GL50evo device settings
   c. it is in Memory mode or directly after a measurement
   d. it is turned on
   e. it is not connected (it will also start advertising after a disconnect from app side)
      i. Since Meter FW version x148 / x149 it is possible to terminate the advertising after disconnect by using the “Custom Service” (section 1.3).